ADVANCED BUSINESS ANALYTICS | FINAL PITCH | OCTOBER 2026

Vietnam Real Estate AVM

An independent fair value for every secondary-market home in Hanoi and HCMC

Team: Advanced Business Analytics capstone group

60% to 64%
apartments, alley houses within ±20%

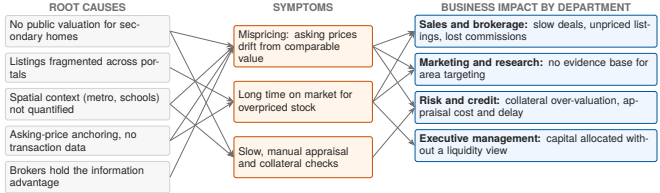
1.5 times
stale rate in the top 10% of risk scores

VND 9.7 bn
net partner value a year, base case

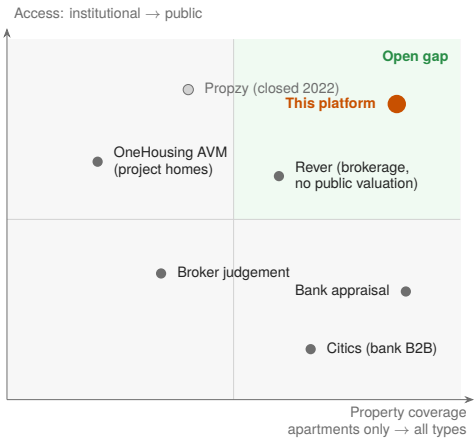
Buyers, brokers and banks price secondary homes with no independent reference

- Homes trade for billions of VND, yet the asking price anchors every negotiation.
- Brokers search comparables by hand; banks wait for manual appraisals.
- The same missing number costs four departments in different ways.

Source: team analysis.



Existing tools cover project apartments or banks; the public, all-type quadrant is empty



Source: company pages and press reports (as reported, not independently verified); team positioning.

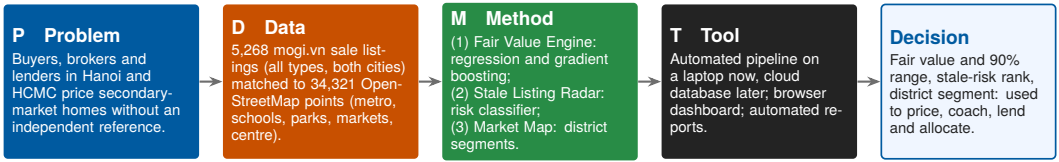
OneHousing: 533,856 homes in 627 projects one month after its 2022 launch; company-reported error below 5%; Techcombank partner.

Citics: valuation used by more than 32 banks; USD 2.1 million raised in 2025; institutions only.

Rever: brokerage, no public valuation. **Propzy:** closed in 2022.

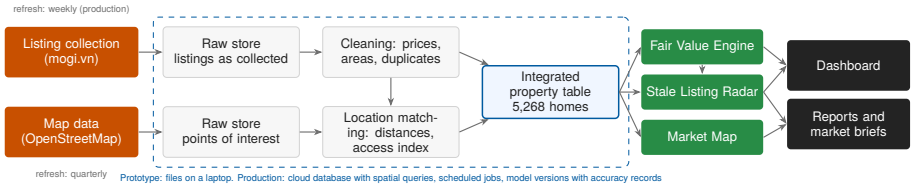
This platform: all property types, published method, error stated by type, listings plus map data.

One picture: problem, data, method and tool lead to three decisions



Source: team design; mogi.vn listings, 15 Oct 2026 (N = 5,268).

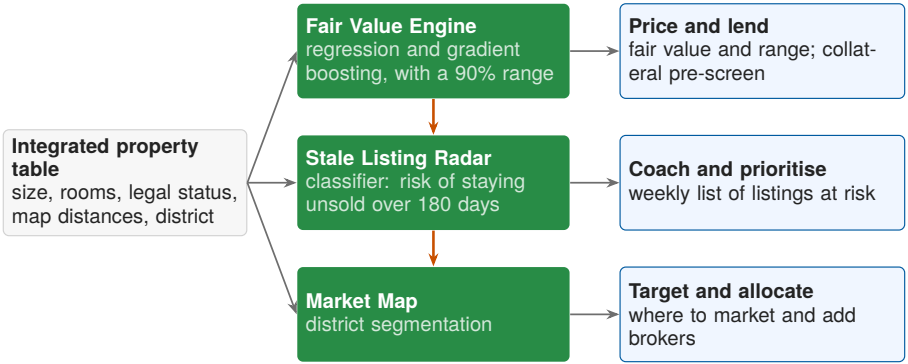
Two sources, one table: 5,268 listings matched to 34,321 map points



A prototype sample sized for a 1.5-hour collection window: 6,000 listings in 46 minutes (7,792 an hour, no blocks); 88% survive cleaning. A scheduled run covers the full market (at least 33,000 listings) in about 5 hours.

Source: mogi.vn listings, 15 Oct 2026 (N = 5,268); OpenStreetMap contributors (ODbL).

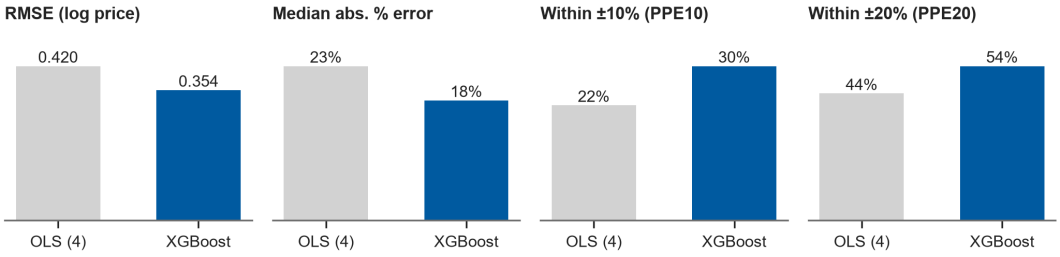
Three linked engines turn one table into three decisions



Every accuracy figure comes from homes the engines had not seen. Listing age is measured from creation, because sellers refresh posted dates (59% re-posted after 30 days).

Source: team design.

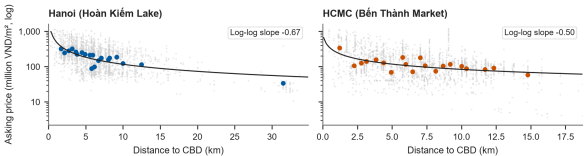
Fair Value Engine: 53.9% of unseen homes within ±20%, better than regression on every measure



- Ready for brokers: alley houses 64%, apartments 60% within ±20%.
- Not yet: street-front houses 42%, land 43%.
- Half of estimates miss by more than 18.1%: always show the 90% range.
- Typical error target of 0.20 (log scale) missed at 0.354.

Source: mogi.vn listings, 15 Oct 2026 (N = 5,268); 1,054 test homes the models had not seen.

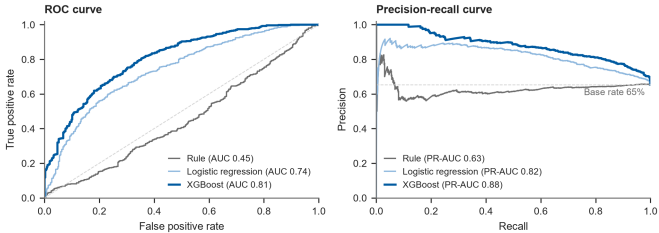
Map data alone delivers 60% of the accuracy gain from knowing the district



- Size, rooms, legal status: R^2 0.714.
- Adding the district cuts the error by 25%.
- Map distances without the district cut it by 15%.
- So the engine can value homes in districts with few listings.

Source: mogi.vn listings, 15 Oct 2026 (N = 5,268); OpenStreetMap contributors (ODbL); regression on log asking price, cross-validated error.

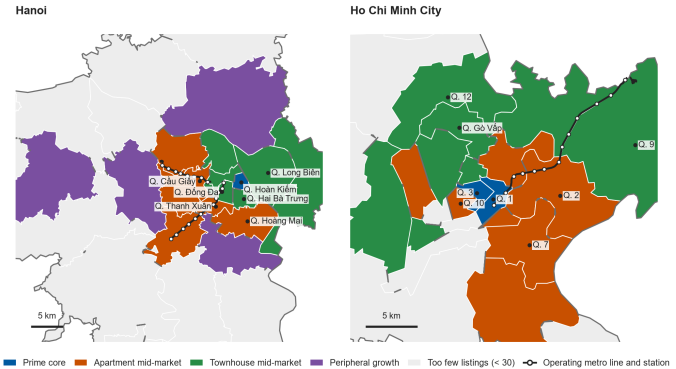
Stale Listing Radar: the top 10% are stale 1.5 times as often as average, and price is not the driver



- AUC 0.81 (0.5 = coin flip); stale = unsold over 180 days.
- Top 10%: 99% stale against 65% on average.
- A price-gap rule is no better than a coin flip; the gap ranks 12 of 26 drivers.
- Use the radar to target effort, the price gap to coach sellers.

Source: mogi.vn listings, 15 Oct 2026 (N = 5,268); 1,054 test listings.

Market Map: four district segments, four different messages



Source: mogi.vn listings, 15 Oct 2026 (N = 5,268); OpenStreetMap contributors (ODbL); district boundaries GADM 4.1.

Prime core: 415 m VND/m^2; defend price.

Apartment mid-market: slowest (364 days); push harder.

Townhouse mid-market: fastest (254 days); volume.

Peripheral growth: land-led, highest stale risk (36%).

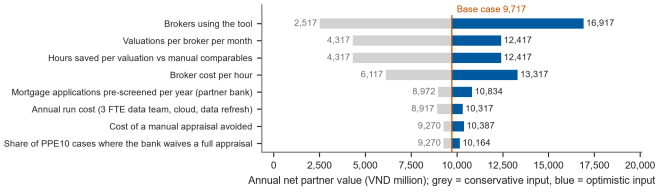
Segments overlap: use them as targeting groups, not hard borders.

Every department gets a decision tool and a KPI

Department	Decision	Evidence	KPI
Sales and brokerage	Which listings to prioritise, which sellers to coach	Top-quarter risk share: 70% in Quận 10 (HCMC); 37% of street-front houses ask >15% above fair value	Days to sale, coached vs not
Marketing and research	Which areas, which message	Four segments differ in price, speed and mix	Leads per listing by segment
Risk and credit	When a fair value can pre-screen collateral	Fair value exceeds the asking price by >20% for 25% of homes; desk review needed	Pre-screens confirmed within ±10%
Executive management	Where to add brokers and offices	Speed vs price matrix by district	Median listing age by segment

Source: team analysis.

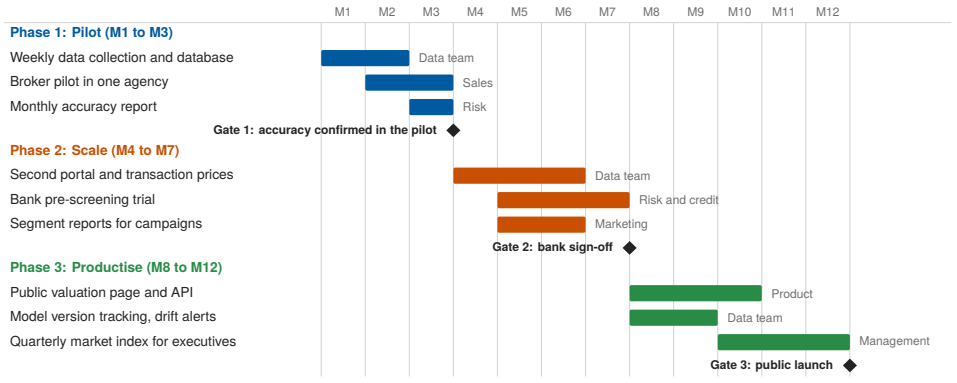
Base case: VND 9.7 bn net partner value a year, 4.4 times the run cost; adoption decides



- Partner value VND 11.9 bn against a run cost of 2.2 bn.
- Range: VND −2.3 to 40.9 bn a year.
- Buyers save VND 10.4 bn (outside the ROI).
- Scenario estimates on labelled assumptions, not measured outcomes.

Source: team assumptions; partner value = broker time saved + bank appraisals avoided.

Twelve months, three phases, a go or no-go gate at each step



Source: team plan.

Risks are known and owned; three asks to start the pilot

Top risks and mitigations

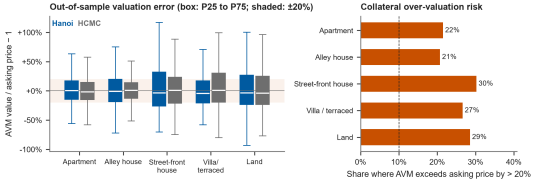
- Asking vs transaction price: calibrate on bank appraisals in Phase 2.
- Noisy stale label: weekly data collection gives true time to sale.
- Site changes: tested page readers, a second portal.
- Publishing valuations: ranges, disclaimers, published method.

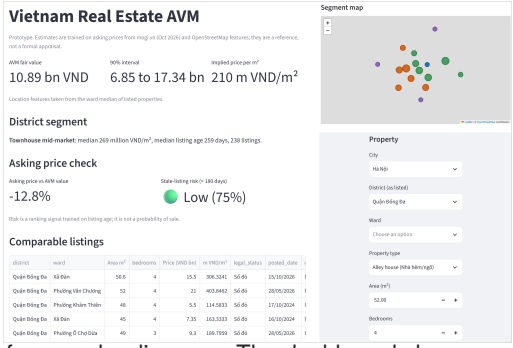
Source: team analysis.

Asks

1. One brokerage partner for a 3-month pilot (apartments, alley houses).
2. Appraisal or notarised prices from a partner bank.
3. Budget for a 3-person data team (VND 2.2 bn a year, base case).

Appendix: a pricing reference for brokers today, not yet a lending tool





Expensive homes tend to be under-valued; monitor before any lending use. The dashboard shows fair value, range, price gap, stale-risk light and comparables.

Source: mogi.vn listings, 15 Oct 2026 (N = 5,268); dashboard prototype.